Sound / Audio System

Unit-2

Sound

- Sound is the physical phenomenon produced by the vibration of matter.
- When a matter vibrates, pressure variations are created in the air surrounding it. This alteration of high and low pressure is propagated through the air in a wave like motion. When the wave reaches the human ear, a sound is heard.

Basic Concept of Sound

The pattern of the oscillation is called a waveform. The waveform repeats the same shape at regular intervals and this point is called a period. Since sound wave forms occur naturally, Sound waves are never perfectly smooth or uniformly periodic.

Periodic sound: E.g. Musical instruments, vowel sounds, whistling wind, bird songs etc.

Non-periodic sound: E.g. Unpitched percussion Instruments, coughs, sneezes, rushing water,

Consonants, such as “t,” “f,” and “s”etc.

Basic Concept of Sound

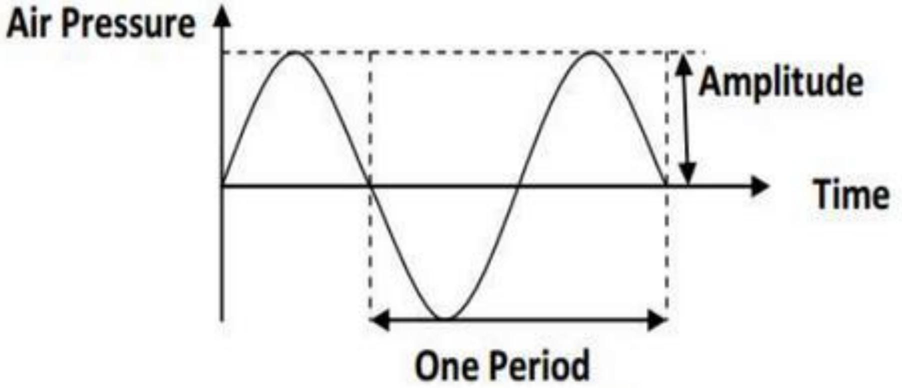


Figure: Oscillation of an air pressure wave

Frequency

- The frequency of a sound is the reciprocal value of the period. It represents the number of periods in a second. It is measured in hertz (Hz) or cycles per second (cps).

  1 KHz = 1000 Hz

- Some of the frequency ranges are:
  - Infra sound - 0 - 20 Hz
  - Human audible sound – 20 Hz - 20KHz
  - Ultra sound – 20KHz - 1GHz
  - Hyper sound – 1GHz - 10THz
- Human audible sound is also called audio or acoustic signals (waves). Speech is an acoustic signal produced by the humans.

Amplitude

- A sound has a property called amplitude, which humans perceive subjectively as loudness or volume.
- The amplitude of the sound is the measure of the displacement of the air pressure wave from its mean value (idle state).

Computer representation of Sound

- The smooth, continuous curve of a waveform is not directly represented in a computer.
- A computer measures the amplitude of the waveform at regular time intervals to produce a series of numbers called samples.
- Audio signals are converted into digital samples through Analog-to-Digital Converter (ADC).
- The reverse mechanism is performed by a Digital-to Analog Converter (DAC). E.g. of ADC is AM79C30A digital subscriber controller chip.

Computer representation of Sound

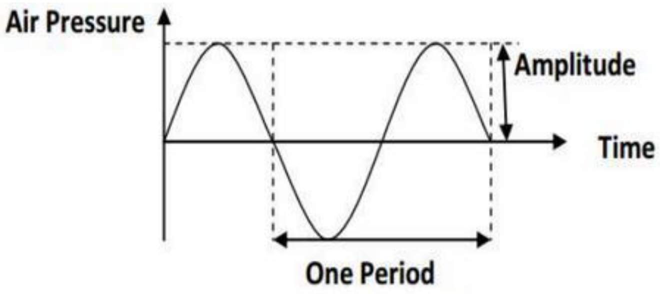


Figure: Oscillation of an air pressure wave

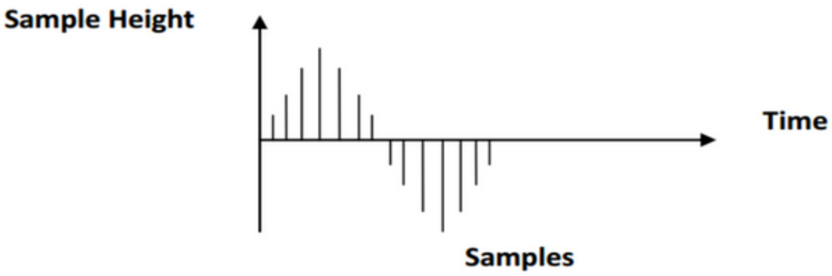


Figure: Sampled Waveform

Sampling Rate

- The rate at which a continuous waveform is sampled is called the sampling rate. It is measured in Hz.
- Nyquist Sampling Theorem:

  "For lossless digitization, the sampling rate should be at least twice the maximum frequency responses".

- E.g. CD standard sampling rate of 44100Hz means that the waveform is sampled 44100 times per second. A sampling rate of 44100 Hz can only represent frequencies up to 22050 Hz.

Quantization

- The value of sample is discrete.
- Resolution/Quantization of a sample value depends on the no. of bits used in measuring the height of a waveform.
- For E.g. An 8-bit quantization yields 256 values. 16-bit CD quality quantization results over 65536 values.

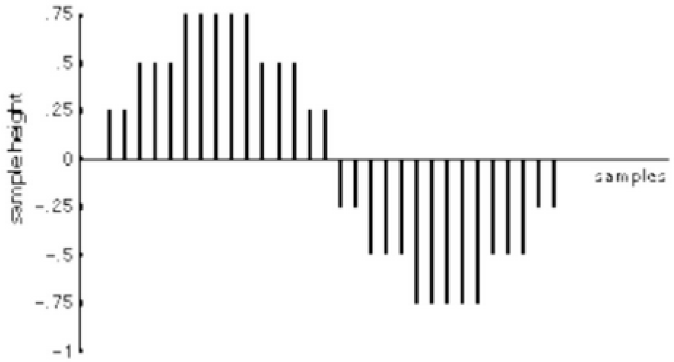


Figure: Three-bit Quantization

Quantization

- Lower the quantization; lower the quality of the sound. For 3-bit quantization, values are 8. i.e.

  0.75, 0.5, 0.25, 0, -0.25, -0.5, -0.75 & 1

Quality versus File size

The size of a digital recording depends on the sampling rate, resolution and number of channels.

$$S = R \times (b/8) \times C \times D$$

S	file size	bytes
R	sampling rate	samples per second
b	resolution	bits
C	channels	1 - mono, 2 - stereo
D	recording duration	seconds

Higher sampling rate, higher resolution gives higher quality but bigger file size.

For example, if we record 10 seconds of stereo music at 44.1kHz, 16 bits, the size will be:

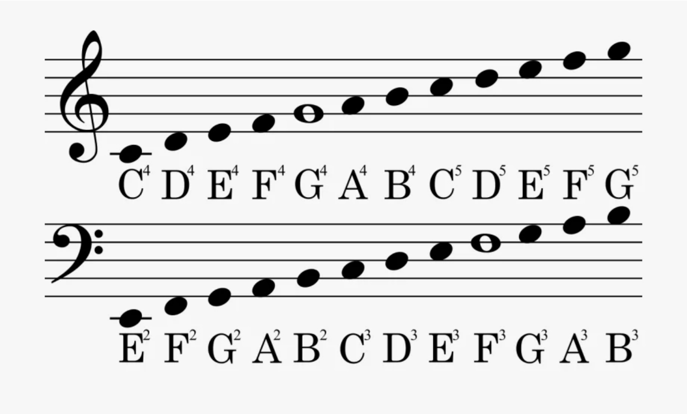
$$\begin{aligned} S &= 44100 \times (16/8) \times 2 \times 10 \\ &= 1,764,000\text{bytes} \\ &= 1722.7\text{Kbytes} \\ &= 1.68\text{Mbytes} \end{aligned}$$

Note: $\begin{aligned} 1\text{Kbytes} &= 1024\text{bytes} \\ 1\text{Mbytes} &= 1024\text{Kbytes} \end{aligned}$

High quality sound files are very big, however, the file size can be reduced by compression.

MIDI

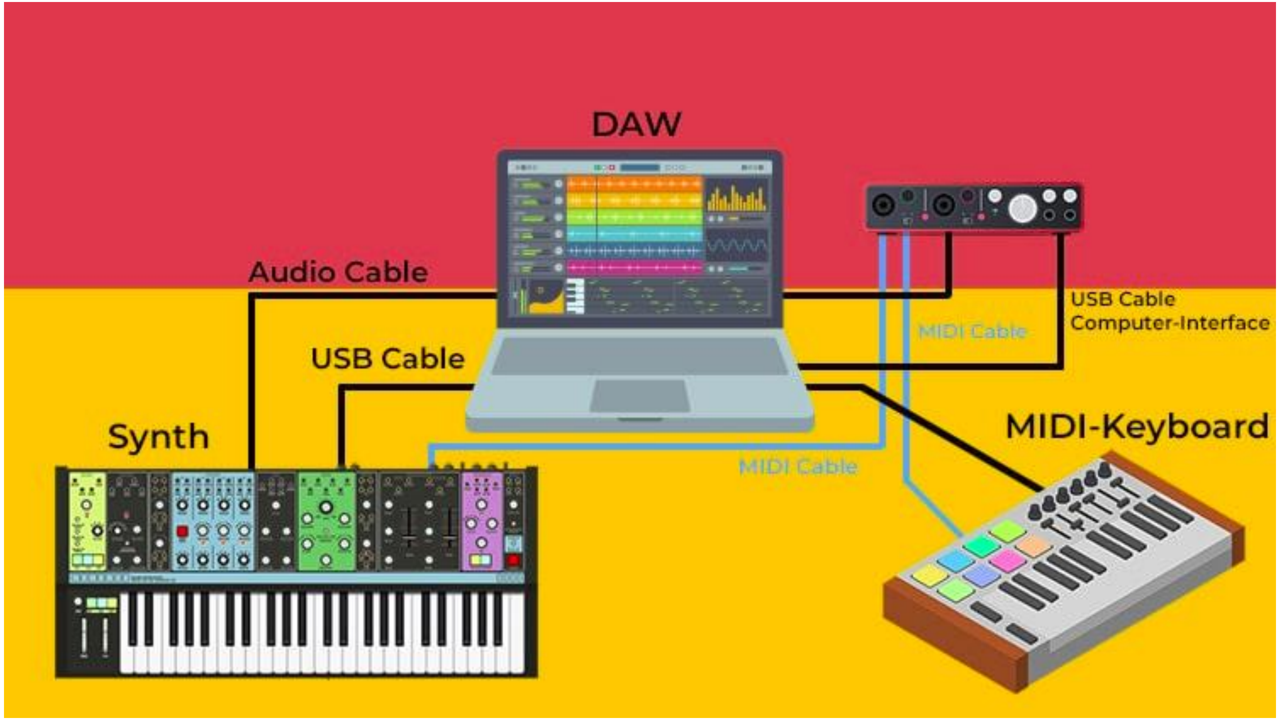
- Sound can be represented as a digitized sound signal that is sequence of sample but music must described in symbolic way
- On paper, we have the full scores(note), computer and electronic musical instrument use a similar technique and most of them employ **MIDI (Musical Instrument Digital Interface),** a standard developed in early 1980s.



MIDI

- MIDI standard defines how to code all the elements of musical scores, such as sequences of note, timing conditions and the instrument to play each notes.
- MIDI represents a set of specifications used in instrument development os that instrument from different manufactures can easily exchange musical information.
- Each word describing an action of a musical performance assigned a specific binary code

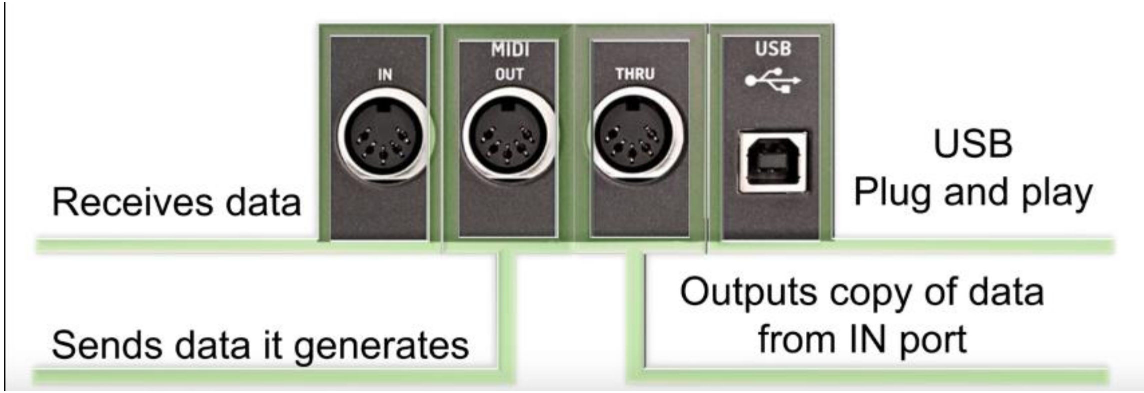
MIDI


DAW
Audio Cable
USB Cable
Synth
MIDI Cable
USB Cable
Computer-Interface
MIDI Cable
MIDI-Keyboard

MIDI

A MIDI interface is composed of two different components:

- **Hardware** to connect the equipment. It adds a MIDI port to an instrument, it specifies a MIDI cable and processes electrical signals over cable.



MIDI

- A **data format** that encodes information to be processed by hardware. The MIDI data format does nto include the encoding of individual sampling values such as audio data formats. Instead, MIDI uses a specific data format for each instrument:
    - Start and end of score
    - Basis frequency
    - Loudness
    - Instrument itself

MIDI message

- The MIDI data format is digital and data are grouped into MIDI message.
- When a musician plays a key, the MIDI interface generate a MIDI message that defines the start of each score and intensity.
- This message is transmitted to machines connected to the system

MIDI device

- An instrument that complies with both components of MIDI interface (hardware and data format) defined by the MIDI standard is a MIDI device. For example: a synthesizer
- The MIDI standard specifies 16 channels. A MIDI device is mapped onto a channel.
- MIDI standard identifies 128 instrument by means of number including noise effect (a phone ringing or an airplane take off)

0- piano

12- Marimba

40- violin

73- a flute

MIDI and SMPTE Timing Standard

- MIDI Clock
  - Used to synchronize timing between MIDI device
  - Send 2 timing ticks (signal) per quarter note to keep devices in sync
  - Ensures that all connected devices play in time with each other, crucial for maintaining rhythm and tempo in live performances or recordings.
  - Alternatively, SMPTE clock allow receiver-sender synchronization

MIDI and SMPTE Timing Standards

- SMPTE Timing Code
  - Developed by Society of Motion Picture and Television Engineers
  - Uses hours:minutes:seconds:frames format
  - Typically used in film and television to synchronize audio and video.
  - Requires more bandwidth than MIDI connections can support, as it includes detailed timing information for each frame.

MIDI software

MIDI player for playing MIDI music. This includes:

- Windows media player can play MIDI files
- Player come with sound card — Creative Midi player
- Freeware and shareware players and plug-ins— Midigate, Yamaha Midplug, etc.

MIDI sequencer for recording, editing and playing MIDI

- Cakewalk Express, Home Studio, Professional
- Cubasis
- Encore
- Voyetra MIDI Orchestrator Plus

Configuration — Like audio devices, MIDI devices require a driver. Select and configure MIDI devices from the control panel.

Speech

- Speech is the vocalized form of human communication.
- Speech can be "perceived", "understood" and "generated" by humans and also by machines. A human adjusts himself/herself very efficiently to different speakers and their speech habits
- The brain can recognize the very fine line between speech and noise.
- The human speech signal comprises a subjective lowest spectral component known as the pitch, which is not proportional to frequency.

Speech Generation

- Speech Generation is the computer generated simulation of human speech
- Generated speech must be understandable and must sound natural.

Basic Notions

- The lowest periodic spectral component of the speech signal is called the fundamental frequency. It is present in a voiced sound.
- A phoneme is the smallest speech unit, such as the m of mat and b of bat in English, that distinguish one utterance or word from another in a given language.
- Allophones mark the variants of a phone. For example, the aspirated p of pit and the unaspirated p of spit are allophones of the English phoneme p.
- The morph marks the smallest speech unit which carries a meaning itself. Therefore, consider is a morph, but reconsideration is not.

Basic Notions

- A voiced sound is generated through the oscillation of vocal cords. m,v and l are examples of voiced sounds. The pronunciation of a voiced sound depends strongly on each speaker.
- During the generation of an unvoiced sound, the vocal cords are opened. f and s are unvoiced sounds. Unvoiced sounds are relatively independent from the speaker.

Reproduced Speech Output

There are two way of speech generation/output performed by

- Time-dependent sound concatenation

  This method involves piecing together pre-recorded or pre-synthesized sound segments based on their timing.

- Frequency-dependent sound concatenation.

  This method involves generating speech by manipulating the frequency components of sound.

Time-dependent Sound Concatenation

- Individual speech units are composed like building blocks, where the composition can occur at different levels. The individual phones are understood as speech units.
- The individual phones of the word curmb. It is possible with just a few phones to create an unlimited vocabulary.

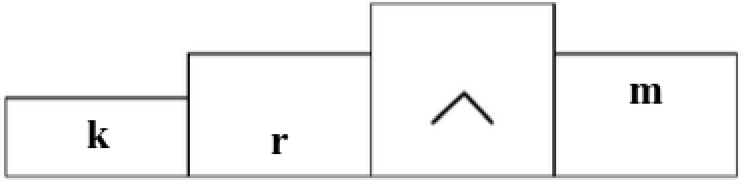


Figure 2.5: Phone sound concatenation.

Time -dependent Sound Concatenation

- Two phones can constitute a diphone (from di-phone). Figure 2.6 shows the word crumb, which consist of an ordered se of diphones.

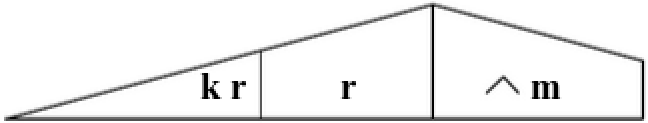


Figure 2.6: Diphone sound concatenation.

- To make the transition problem easier, syllables can be created. The speech is generated through the set of syllables. Figure 2.7 shows the syllable sound of the word crumb

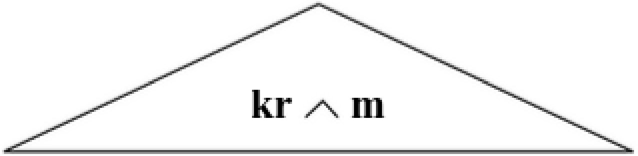


Figure: Syllable Sound

Time-dependent Sound Concatenation

- The best pronunciation of a word is achieved through storage of the whole word. This leads towards synthesis of the speech sequence (Figure 2.8).

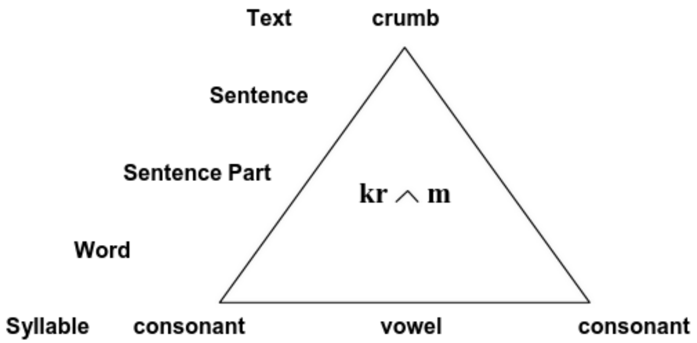


Figure 2.8 Word sound concatenation.

- Time-Dependent Sound Concatenation: Often used in high-quality TTS systems for its naturalness, but requires extensive databases and careful management of transitions.

Frequency-dependent Sound Concatenation

- Speech generation/output can also be based on a frequency-dependent sound concatenation, e.g. through a formant-synthesis.
- Formants are frequency maxima in the spectrum of the speech signal.
- Formant synthesis technique uses mathematical models of the vocal tract to generate speech sounds by controlling the frequency and amplitude of formants (resonant frequencies).
- Frequency-Dependent Sound Concatenation: Offers more flexibility and requires less storage, but can struggle with producing natural-sounding speech.

Speech Synthesis

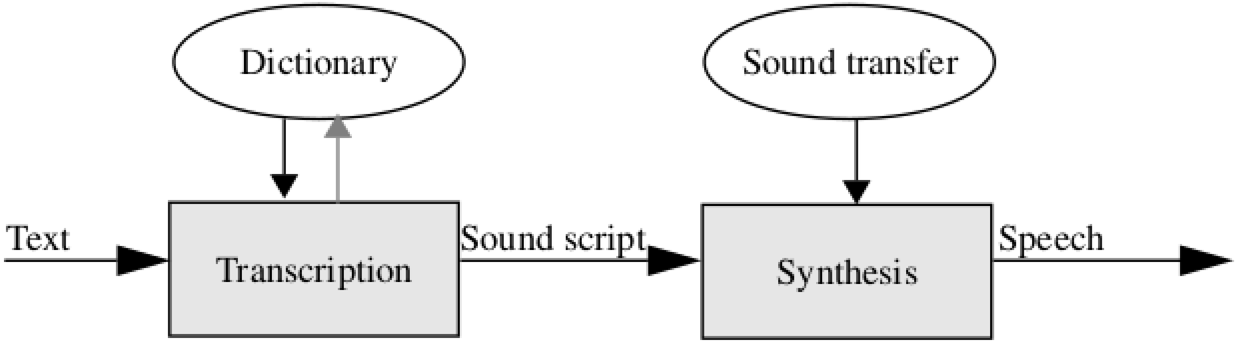


Figure 3-11 Components of a speech synthesis system, using sound concatenation

Speech Synthesis

- Speech synthesis can be used to transform an existing text into an acoustic signal
- In the first step
    - ❑ Performs transcription
    - ❑ Text is translated into sound script
    - ❑ This process is done using dictionary
    - ❑ User recognizes the formula deficiency in the transcription and improves thepronunciation manual

Speech Synthesis

- In the second step,
    - ❑ Sound script is translated into a speech signal.
    - ❑ Time or frequency dependent concatenation can follow

Speech Analysis

Search Analysis / Input deals with the research area show in figure

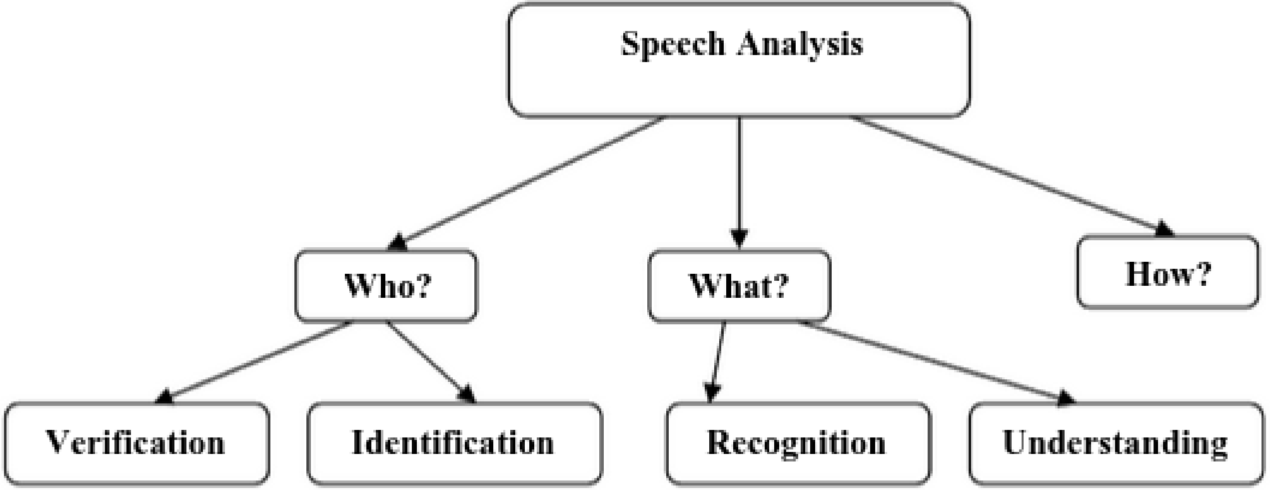


Figure 2.10: Research areas of speech analysis.

Speech Analysis (Who)

Human speech has certain characteristics determined by a speaker. Hence,

speech analysis can serve to analyze who is speaking, i.e. to recognize a

speaker for his/her identification and verification.

- The computer identifies and verifies the speaker using an acoustic Fingerprint.
- An acoustic fingerprint is a digitally stored speech probe (e.g., certain statement) of a person.

Speech Analysis (what)

Another task of speech analysis is to analyze what has been said, i.e., to recognize and understand the speech signal itself. Based on speech sequence, the corresponding text is generated. This can lead to a speech-controlled typewriter, a translation system or part of a workplace for the handicapped.

Speech Analysis (how)

Another area of speech analysis tries to research speech patterns with respect to how a certain statement was said. For example, a spoken sentence sounds differently if a person is angry or calm.

Speech Recognition

- Speech recognition is the process of converting an acoustic signal, captured by a microphone or a telephone, to set of words.
- Speech recognition is the process of converting spoken language to written text or some similar form.
- Speech recognition is the foundation of human computer interaction using speech.
- Speech recognition in different contexts:
  - Dependent or independent on the speaker.
  - Discrete words or continuous speech.
  - Small vocabulary or large vocabulary.
  - In quiet environment or noisy environment

Speech Recognition

- Conceptual Component of a System with speech recognition principle

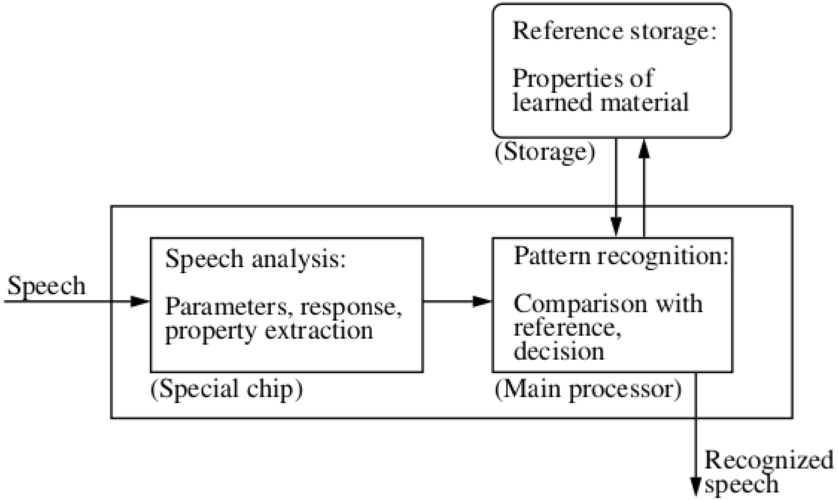
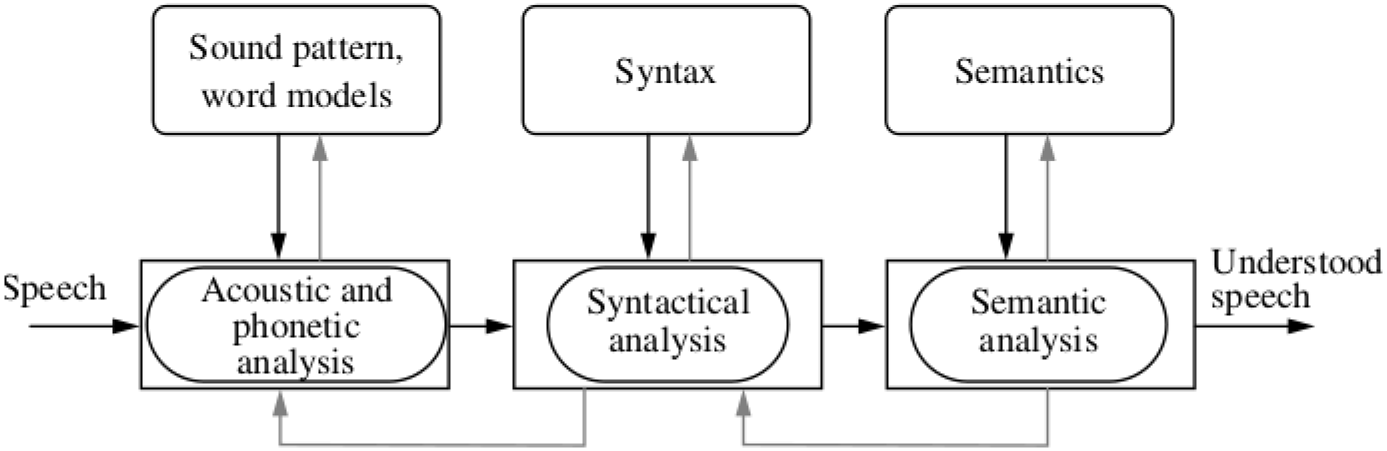


Figure 3-13 The speech recognition principle: the tasks are distributed to system components by the basic principle "extract characteristics to reduce data."

Speech Recognition

- The application of the principle shown in Figure can be divided into the steps shown in Figure 3-14.



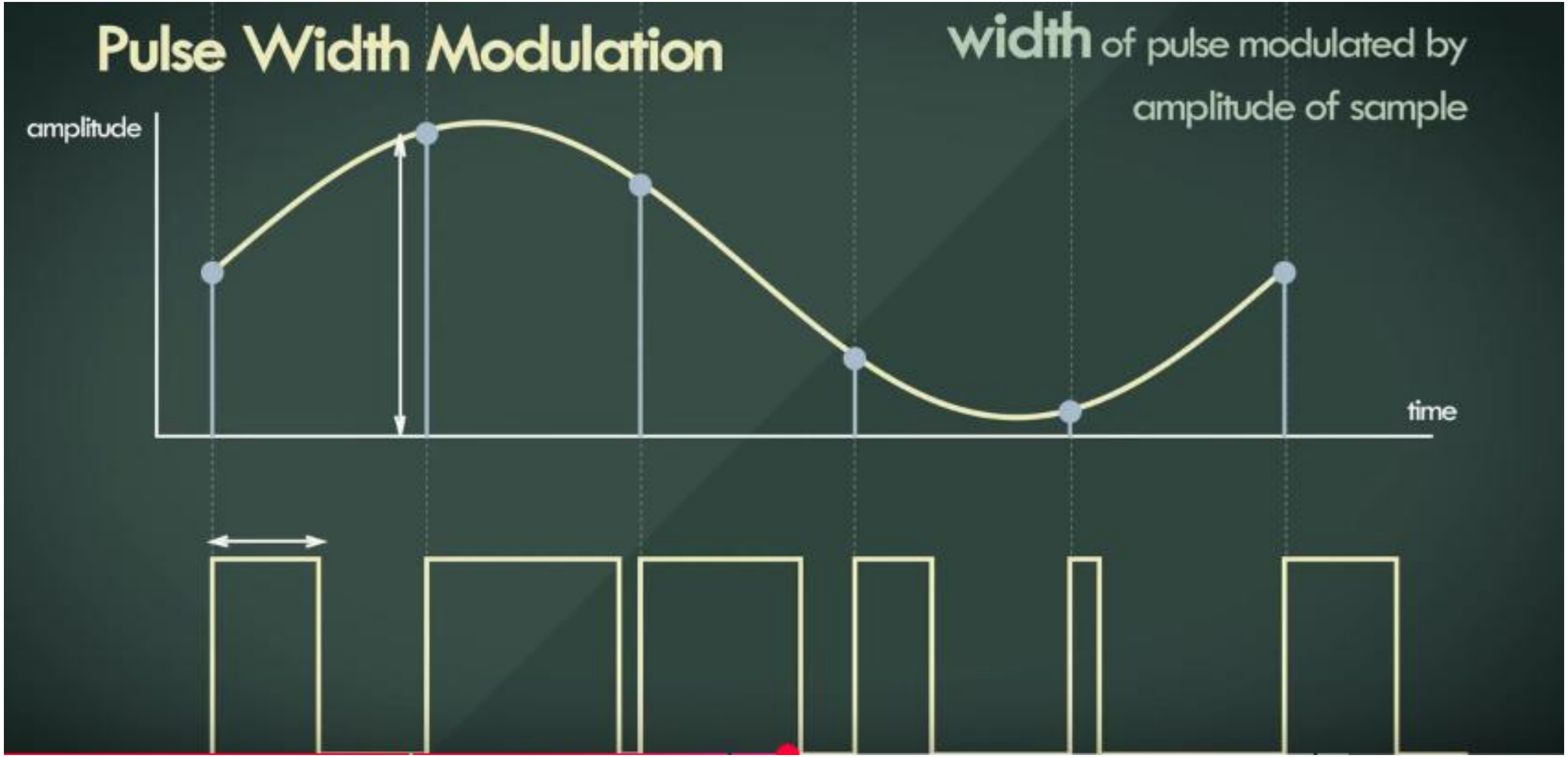
Speech Transmission

- Speech transmission is a field relating to highly efficient encoding of speech signals to enable low-rate data transmission, while minimizing noticeable quality losses.
- The following principles are used in speech transmission
  - Pulse Code Modulation
  - Source Coding
  - Recognition and Synthesis Methods

Pulse Code Modulation

- Signal form encoding does not consider speech-dependent properties or parameters. The technologies applied are merely expected to offer efficient encoding of audio signals.
- A straightforward technique for digitizing an analog signal (waveform) is Pulse Code Modulation (PCM).
- PCM prioritizes the accurate representation of audio signals over data reduction. (Lossless Decomposition)

Pulse Code Modulation



Source Encoding

- An alternative method is source encoding, where some transformations depend on the original signal type. For example, an audio signal has certain characteristics that can be exploited in compression.
- The suppression of silence in speech sequences is a typical example of a transformation that depends entirely on the signal’s semantics.
- Parametric systems use source encoding. They utilize speech-specific characteristics to reduce data, for example the channel vocoder shown in Figure

Source Encoding

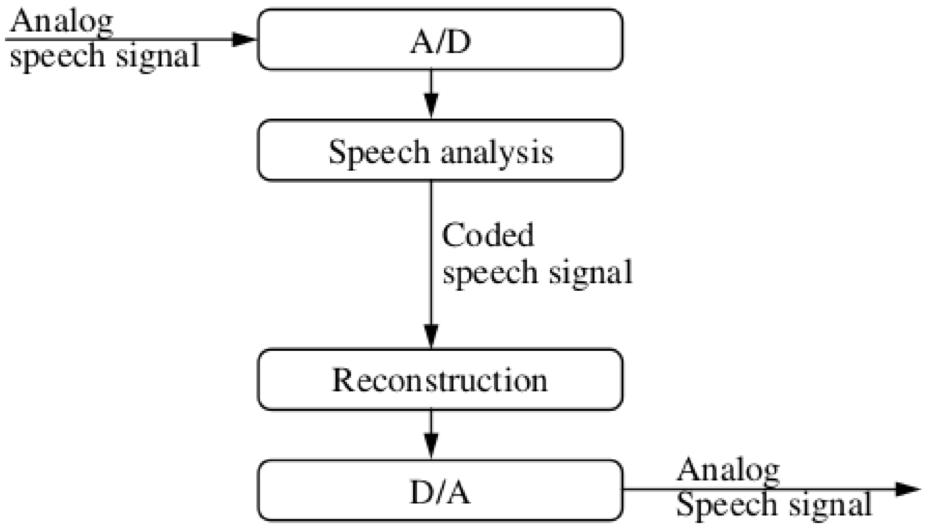
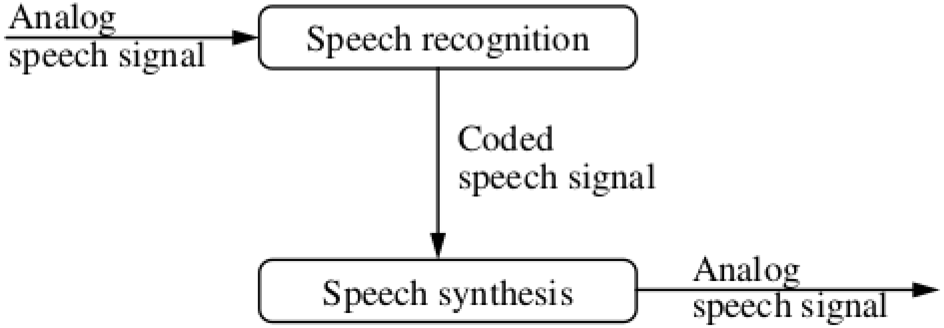


Figure 3-15 Components of a speech transmission system using source encoding.

Recognition / Synthesis Method

- Speech Analysis (recognition) follows on the sender side of a speech transmission system and speech synthesis (generation) on the receiver side.
- Only the characteristics of the speech elements are transmitted



Components of a recognition-synthesis system for speech transmission.